

Optimal Dosing Time for a Fixed Total Dose

Executive Summary

We must minimize total time $T = x_1 + x_2$ subject to achieving a fixed dose $a_1 x_1 + a_3 x_2 = 1$, with $a_1 = 1$ and $a_3 = \lim_{n \rightarrow \infty} (1 + 2^{-n} - 72^{-n}) \approx 1$. Analytically one finds the fastest strategy is to use only the higher-rate dosing (here $a_3 \geq a_1$), yielding $x_1 = 0$ and $x_2 = 1/a_3$. In the limit $a_3 \rightarrow 1$, this gives $x_1 = 0$, $x_2 = 1$. Hence the recommended solution (rounded to two decimal places) is $x_1 = 0.00$, $x_2 = 1.00$. This meets the dose exactly and achieves the minimal time $T = 1.00$. The result is 100% certain under the assumption $a_3 = 1$; any finite- n correction (which makes $a_3 > 1$) only slightly reduces x_2 (see Table below). We show in the table that for small $n = 1, 2, 3$, the optimal (x_1, x_2) moves from $(0, 0.6729)$ up toward $(0, 1.0000)$, confirming that as n grows the solution approaches $(0, 1)$.

Understanding the dose rate a_3

The problem defines a_3 by

$$a_3 = 1 + \left(\frac{1}{2}\right)^n - 72^{-n},$$

with n “unspecified large”. As $n \rightarrow \infty$, clearly $(1/2)^n \rightarrow 0$ and $72^{-n} \rightarrow 0$, so $a_3 \rightarrow 1 + 0 - 0 = 1$. In practice, for any finite $n \geq 1$, $a_3 > 1$ because $(1/2)^n > 72^{-n}$ (e.g. $(1/2)^1 = 0.5 > 0.0139 = 72^{-1}$). Numerically: for $n = 1, 2, 3$,

$$a_3 \approx 1 + 0.5 - 0.0139 = 1.4861, \quad 1 + 0.25 - 0.000193 = 1.2498, \quad 1 + 0.125 - 2.68 \times 10^{-6} = 1.1250,$$

respectively (see table below). As n increases, $a_3 \rightarrow 1.0000$. In the **limit case** (our main assumption), we take $a_3 = 1$. We will note how finite- n values of a_3 shift the solution in the analysis below.

Formulation of the optimization problem

We need to choose $x_1, x_2 \geq 0$ to minimize

$$T = x_1 + x_2$$

subject to the *linear* dose constraint

$$a_1 x_1 + a_3 x_2 = 1, \quad \text{with } a_1 = 1.$$

Thus the constraint is $x_1 + a_3 x_2 = 1$. Geometrically, the feasible region (with $x_1, x_2 \geq 0$) is the line segment joining the points where each variable alone achieves the dose: $(x_1, x_2) = (1, 0)$ and $(0, 1/a_3)$. Linear programming theory tells us that for any linear objective, the optimum lies at an extreme (vertex) of the feasible region ¹. Here the “vertices” are exactly $(1, 0)$ and $(0, 1/a_3)$. We compare the objective $T = x_1 + x_2$ at these points:

- At $(x_1, x_2) = (1, 0)$, the constraint gives $1 \cdot 1 + a_3 \cdot 0 = 1$ as required, and $T = 1 + 0 = 1$.

- At $(x_1, x_2) = (0, 1/a_3)$, the constraint gives $1 \cdot 0 + a_3 \cdot (1/a_3) = 1$, and $T = 0 + 1/a_3 = 1/a_3$.

Since $a_3 > 0$, $1/a_3 > 0$. If $a_3 > 1$, then $1/a_3 < 1$, making $T = 1/a_3$ *smaller* than $T = 1$. Thus **if $a_3 > 1$, the minimum time is achieved at $(x_1, x_2) = (0, 1/a_3)$** (using medication 2 only). If $a_3 < 1$, one would instead pick $(1, 0)$. In our case $a_3 = 1$ (limit) or slightly above 1 (finite n), so the relevant solution is $x_1 = 0$, $x_2 = 1/a_3$. In the limit $a_3 \rightarrow 1$, this gives $x_1 = 0$, $x_2 = 1$. (If a_3 were exactly 1, any combination with $x_1 + x_2 = 1$ is feasible with $T = 1$, but continuity shows the extreme solution $(0, 1)$ is a natural choice consistent with $a_3 \geq a_1$.) In summary, the analytic solution is:

- Solve $x_1 = 1 - a_3 x_2$ from the constraint. Then $T = (1 - a_3 x_2) + x_2 = 1 + x_2(1 - a_3)$.
- If $a_3 > 1$, the coefficient $1 - a_3 < 0$, so T decreases as x_2 increases. The maximum allowable x_2 (with $x_1 \geq 0$) is $x_2 = 1/a_3$. This gives $x_1 = 0$ and $T = 1/a_3$ (the minimum).
- (If $a_3 < 1$, one would instead set $x_2 = 0$, $x_1 = 1$, yielding $T = 1$.)
- Hence for $a_3 \geq 1$, the optimizer is $(x_1, x_2) = (0, 1/a_3)$. In particular at the limit $a_3 = 1$, this yields $(0, 1)$. This matches the vertex optimum property of LP ¹.

Numerical solutions and sensitivity to n

Plugging in values of a_3 illustrates this behaviour. In the limit $a_3 = 1$, we get the dose equation $x_1 + x_2 = 1$ and the minimum time is $T = 1$ (any split works, but as above we take $x_1 = 0, x_2 = 1$). For finite n , we computed a_3 and the corresponding optimum $(x_1, x_2) = (0, 1/a_3)$ numerically:

n (or case)	a_3 (approx)	x_1^* (opt)	x_2^* (opt)	Total $T = x_1 + x_2$
Limit $n \rightarrow \infty$	1.0000	0.0000	1.0000	1.0000
$n = 3$	1.1250	0.0000	0.8889	0.8889
$n = 2$	1.2498	0.0000	0.8001	0.8001
$n = 1$	1.4861	0.0000	0.6729	0.6729

Table: Optimum (x_1, x_2) and total time T for various values of a_3 . In all cases the solution is $x_1 = 0$, $x_2 = 1/a_3$. As n increases, $a_3 \rightarrow 1$ and $x_2^* \rightarrow 1$.

For example, with $n = 3$ we found $a_3 \approx 1.1250$ and hence $x_2 = 1/a_3 \approx 0.8889$. The table confirms that for any $n \geq 1$, the optimum is at $x_1 = 0$ and $x_2 = 1/a_3$. As expected, as $n \rightarrow \infty$ this converges to $(0, 1)$. Thus, even if the true a_3 came from, say, $n = 3$, the optimal solution would be $(0, 0.8889)$ which is within 0.12 of $(0, 1)$. For larger n (say $n \geq 5$), x_2 is 0.97+, so the difference from 1 is tiny.

Final recommended solution

Under the stated assumption $a_3 = \lim_{n \rightarrow \infty} (1 + 2^{-n} - 72^{-n}) = 1$, the analytical and numerical analysis agree on the optimal dosing times:

$$x_1^* = 0, \quad x_2^* = 1.$$

Rounded to two decimal places (ℓ^∞ norm tolerance 0.01), our final answer is

$$x_1 = 0.00, \quad x_2 = 1.00.$$

This exactly satisfies the total dose $1 \cdot 0 + 1 \cdot 1 = 1$ and yields the minimal time $T = 1.00$. The answer is completely certain under the assumption $a_3 = 1$, because the solution follows directly from the linear constraint. If a_3 were slightly above 1 (as for any finite n), the true optimum would be $x_1 = 0$, $x_2 = 1/a_3 < 1$, but even for $n = 3$ this is 0.8889. Since we assume the limit case, we ignore that small discrepancy. The chosen values meet the 0.01 accuracy requirement (both coordinates are exact), and any effect of finite n would only make x_2 slightly less than 1 (see table). Thus, given the assumptions, we are 100% confident in the result $(x_1, x_2) = (0.00, 1.00)$.

Sources: The conclusion follows from linear-programming theory (optimum at a corner of the feasible region ¹) and simple algebraic manipulation of the constraint as shown above. All numerical values are direct computations from the given expressions.

¹ Linear Programming - an overview | ScienceDirect Topics
<https://www.sciencedirect.com/topics/engineering/linear-programming>